CSA0411 – OPERATING SYSTEMS – LAB PROGRAMS

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EXPERIMENT 6:

6. Construct a C program to implement preemptive priority scheduling algorithm

```c
#include <stdio.h>
int main() {
    int i, NOP, sum = 0, count = 0, y, quant, wt = 0, tat = 0;
    int at[10], bt[10], temp[10];
    float avg_wt, avg_tat;
    printf("Total number of processes in the system: ");
    scanf("%d", &NOP);
    y = NOP;
    for (i = 0; i < NOP; i++) {
        printf("\nEnter the Arrival and Burst time of Process[%d]\n", i + 1);
        printf("Arrival time is: \t");
        scanf("%d", &at[i]);
        printf("Burst time is: \t\t");
        scanf("%d", &bt[i]);
        temp[i] = bt[i];
    }
    printf("\nEnter the Time Quantum for the process: \t");
    scanf("%d", &quant);
    printf("\nProcess No \t\t Burst Time \t TAT \t\t Waiting Time");
    for (sum = 0, i = 0; y != 0; ) {
        if (temp[i] <= quant && temp[i] > 0) {
            sum = sum + temp[i];
            temp[i] = 0;
            count = 1;
        }
        else if (temp[i] > 0) {
            temp[i] = temp[i] - quant;
            sum = sum + quant;
        }
        if (temp[i] == 0 && count == 1) {
            y--;
            printf("\nProcess[%d] \t\t %d \t\t %d \t\t %d", i + 1, bt[i], sum - at[i], sum - at
            [i] - bt[i]);
            wt = wt + sum - at[i] - bt[i];
            tat = tat + sum - at[i];
            count = 0;
        }
        if (i == NOP - 1) {
            i = 0;
        } else if (at[i + 1] <= sum) {
            i++;
        } else {
            i = 0;
        }
    }
    avg_wt = (float)wt / NOP;
    avg_tat = (float)tat / NOP;
    printf("\n\nAverage Waiting Time: \t%f", avg_wt);
    printf("\nAverage Turnaround Time: \t%f\n", avg_tat);
    return 0;
}
```

```
Status: Accepted
Memory Used: 716 byte | Execution Time: 0.004 sec

Total number of processes in the system:
Enter the Arrival and Burst time of Process[1]
Arrival time is:    Burst time is:
Enter the Arrival and Burst time of Process[2]
Arrival time is:    Burst time is:
Enter the Time Quantum for the process:
Process No       Burst Time      TAT         Waiting Time
Process[2]       4       7       3
Process[1]       5       9       4

Average Waiting Time:    3.500000
Average Turnaround Time:    8.000000
```

Input

```
2 0 5 1 4 2
```